

Modified Pfirrmann Grading System for Lumbar Intervertebral Disc Degeneration

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Inventory Category:	Clinical grading / reference standard
Comparability / Priority:	Background Priority: Important

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Clinical grading standard Sagittal T2-weighted MRI
Dataset & Cohort:	nan (Sample: nan)
Disease / Target:	Disc degeneration
Model / AI Method:	Modified manual grading
Evaluation Metrics:	nan
Key Finding / Relevance:	nan
Thesis Context (Ch 2):	Shows that degeneration grading is not limited to one five-grade scheme.

Abstract / Summary

Study designA reliability study was conducted. OBJECTIVE.: To modify a grading system for lumbar disc degeneration and to test the reliability of this modified grading system.Summary of background dataThe 5-level Pfirrmann grading system for disc degeneration did not prove discriminatory when used to assess disc degeneration in the elderly spine. Such discriminatory power is necessary to test the association between other variables and severity of disc degeneration.MethodsAn 8-level modified grading system for lumbar disc degeneration was developed including a description of the changes expected for each grade and a 24-image reference panel. The reliability of the modified grading system was tested on 260 lumbar intervertebral discs in 52 subjects (26 men, 26 female) with a mean age of 73 years (range, 67-83 years). All examinations were analyzed independently by 3 readers. Intraobserver and interobserver reliabilities were assessed by calculating weighted kappa statistics.ResultsOn average, for all 3 readers, 0.39% of the 260 discs were classified as Grade 2, 22% were classified as Grade 3, 21.5% were classified as Grade 4, 25.3% were classified as Grade 5, 19.1% were classified as Grade 6, 7.1% were classified as Grade 7, and 4.8% were classified as Grade 8. Intraobserver agreement was excellent (weighted kappa range, 0.79-0.91) with substantial interobserver agreement (weighted kappa range, 0.65-0.67). Complete intraobserver agreement was obtained, on average, in 85% of all discs with 84% of disagreement being as a result of a 1 grade difference. Complete interobserver agreement was obtained, on average, in 66% of all discs with 91% of disagreement being as a result of a 1 grade difference.ConclusionThe modified Pfirrmann grading system is useful at discriminating severity of disc degeneration in elderly subjects. The system can be applied with good intra- and interobserver agreement.